

Video Streaming Systems

Video streaming apps

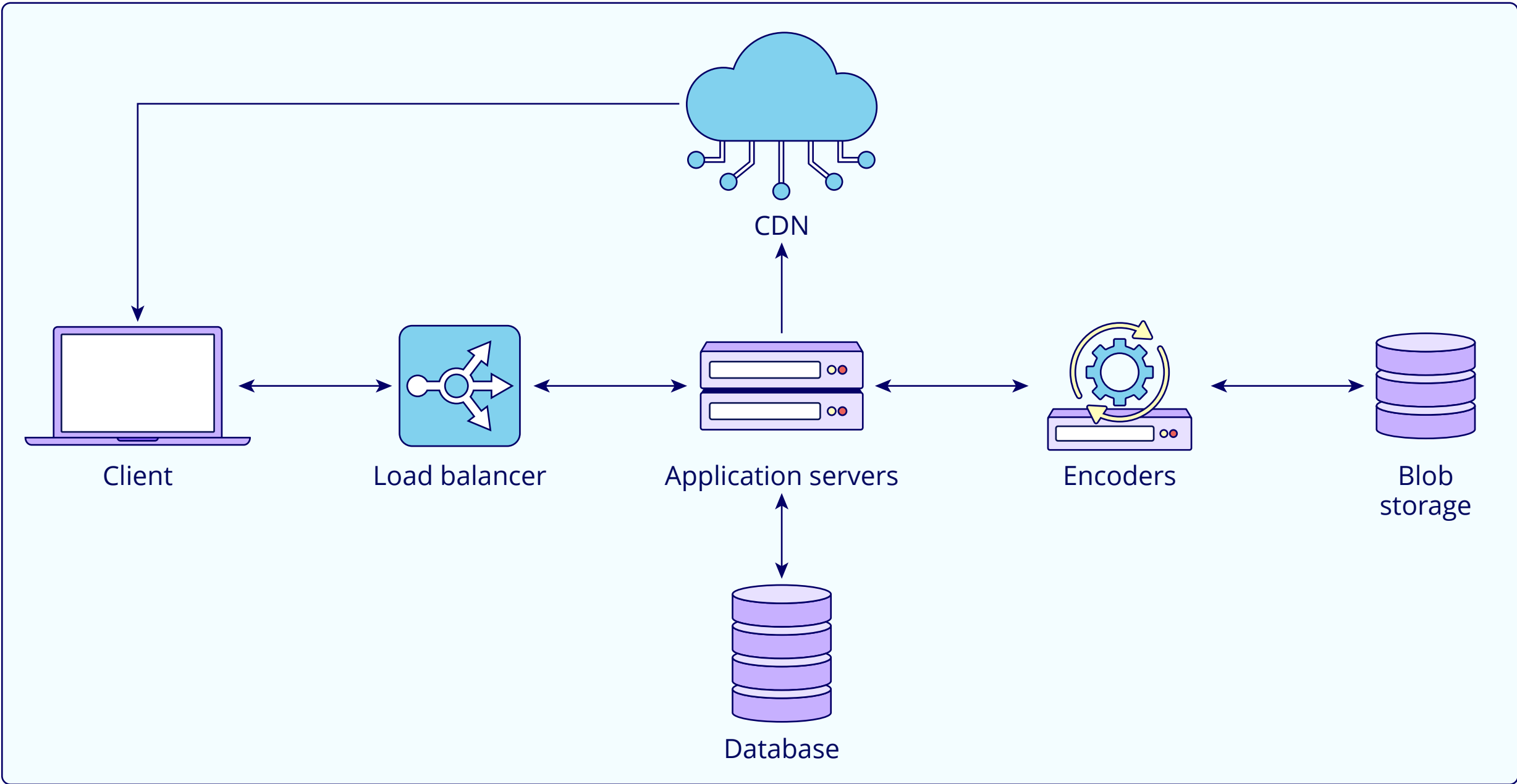


Functional requirements:

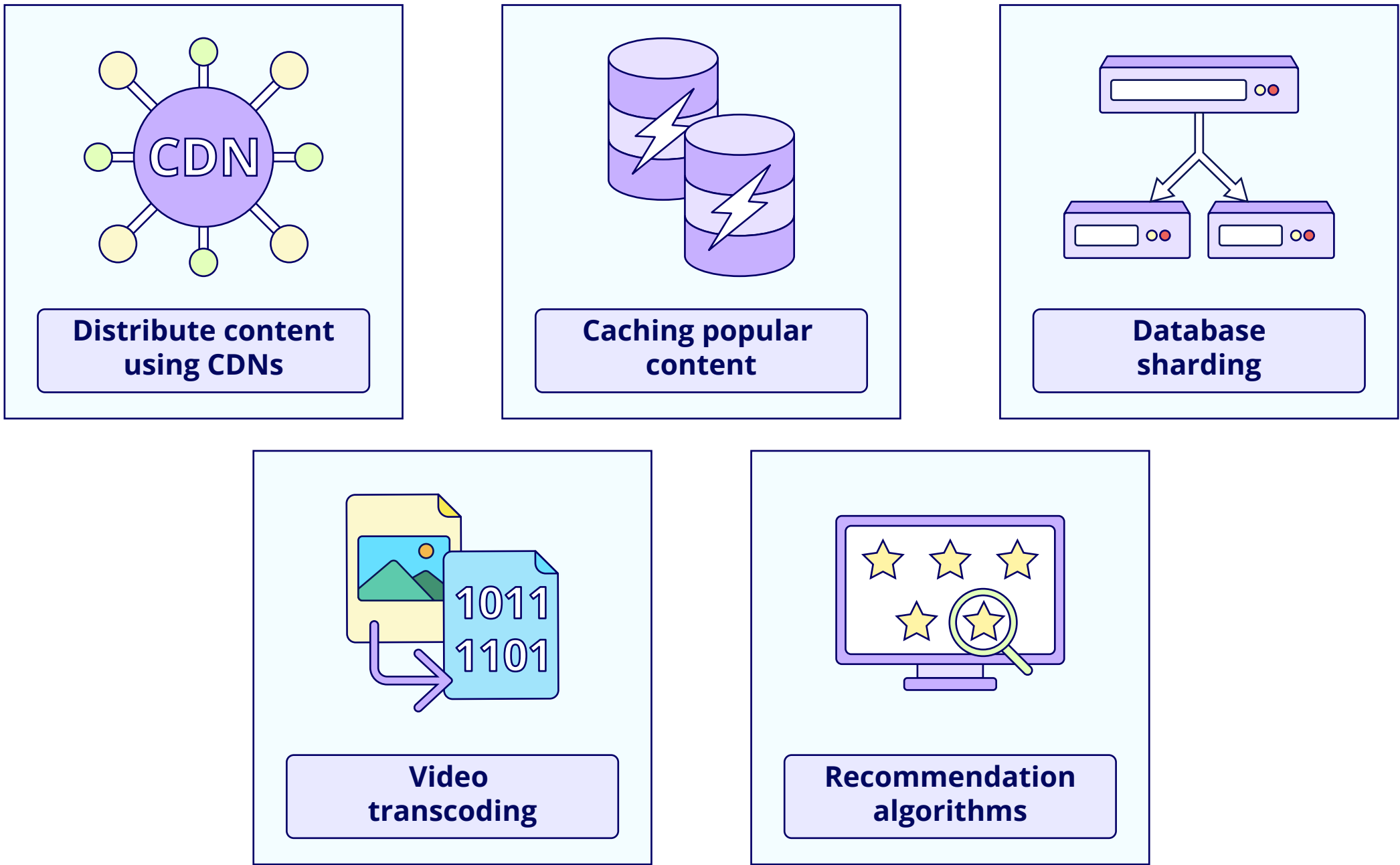
- Stream a video
- Upload a video
- Search a video
- Like or dislike a video
- Provide comments on a video

Nonfunctional requirements:

- High availability
- Low latency
- Scalability
- Reliability



Key techniques to leverage



Real-Time Communication Applications

Communication apps



WhatsApp



Messenger



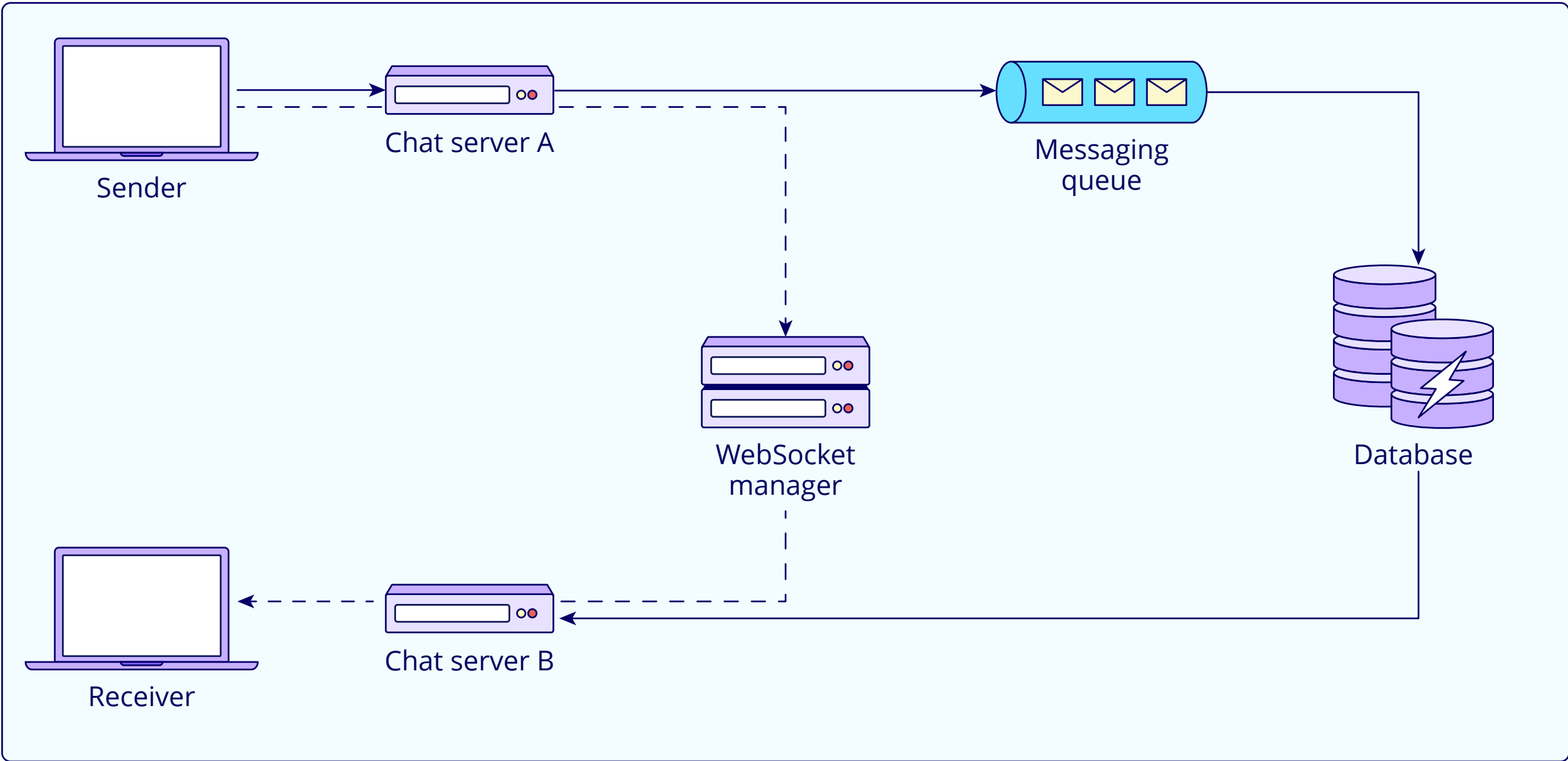
Discord

Functional requirements:

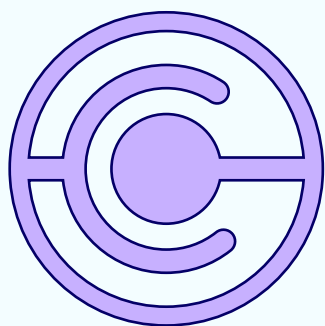
- Real-time communication
- Sharing of media content
- Chat storage
- Notifications
- Group chat

Nonfunctional requirements:

- Low latency
- Scalability
- Availability
- Consistency
- Security



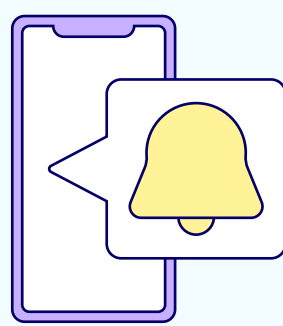
Key techniques to leverage



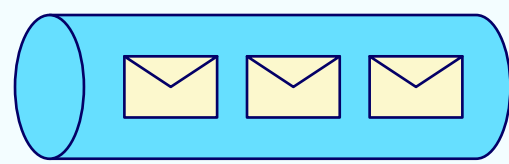
WebSockets



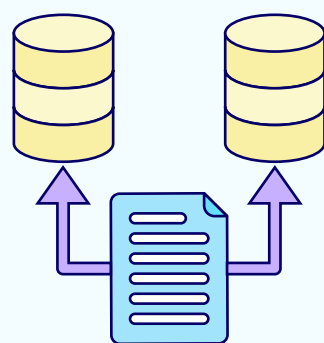
Server-sent
events



Push
notifications



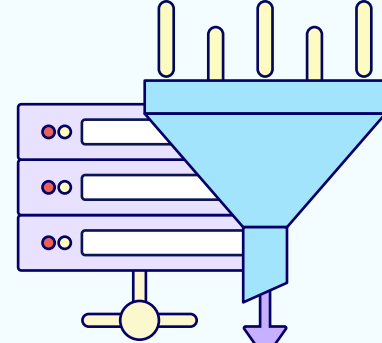
Database
sharding



Database
replication



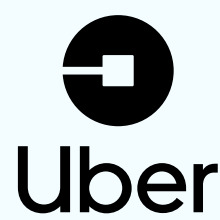
End-to-end
encryption (E2EE)



Rate
limiting

Ride-Hailing Systems

Ride-hailing apps

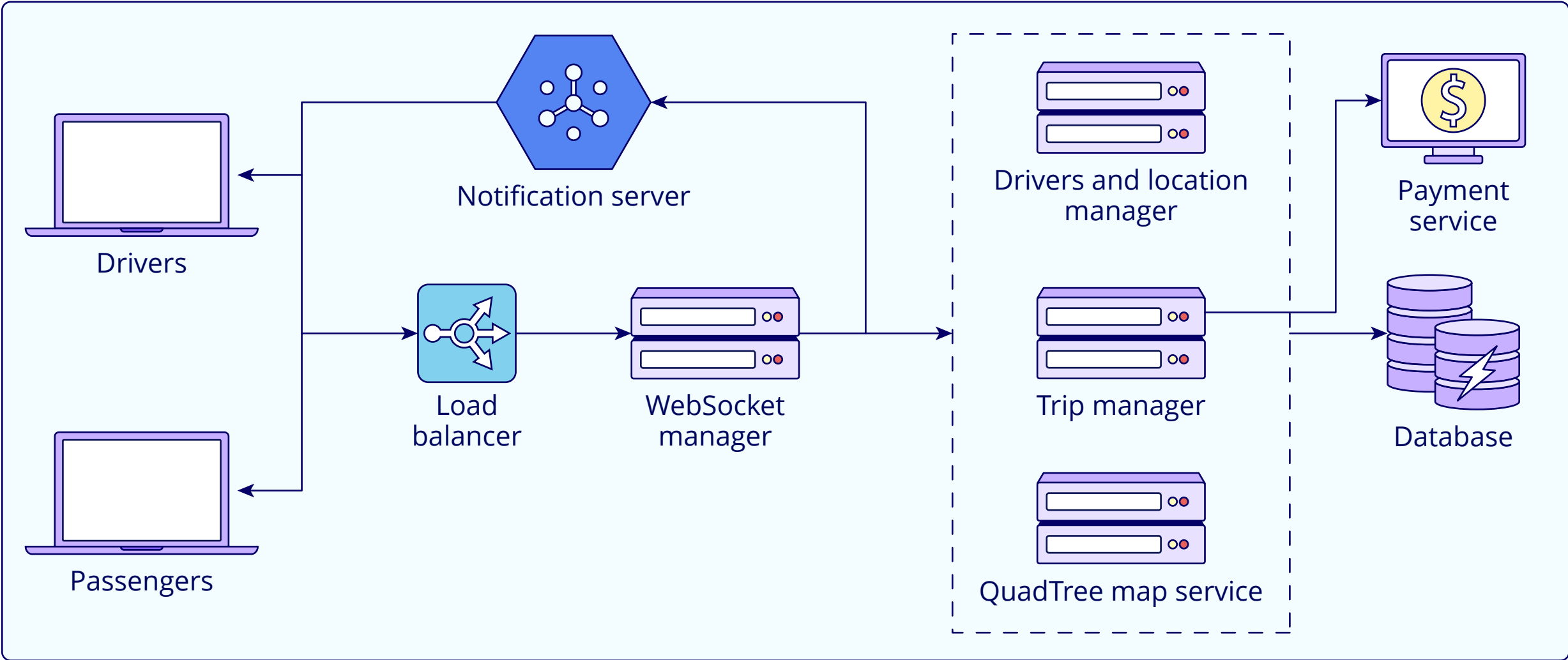


Functional requirements:

- Locate nearby drivers
- Ride request
- Drive location live update
- Payment management
- Driver arrival time
- Pickup confirmation
- Trip updates
- End the trip

Nonfunctional requirements:

- Scalability
- Availability
- Reliability
- Consistency



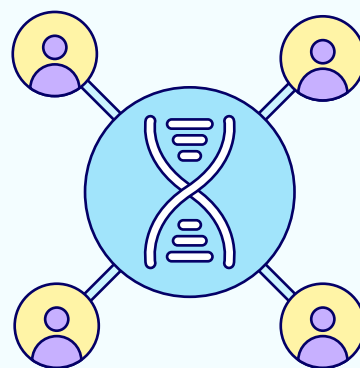
Key techniques to leverage



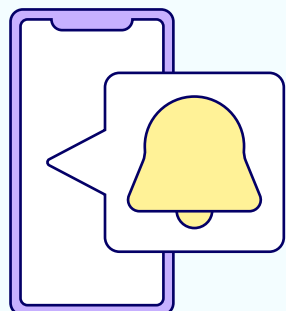
Geolocation services (GPS)



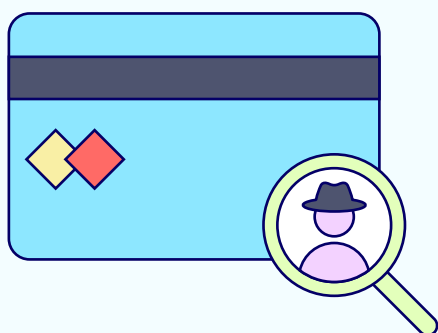
Geospatial databases



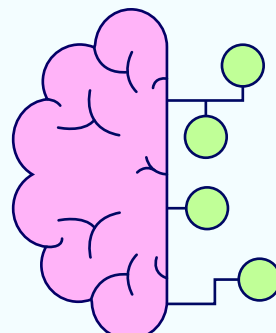
Matching algorithms



Push notifications



Fraud detection



Pricing using machine learning

Newsfeed-Based Social Networks

Social network apps

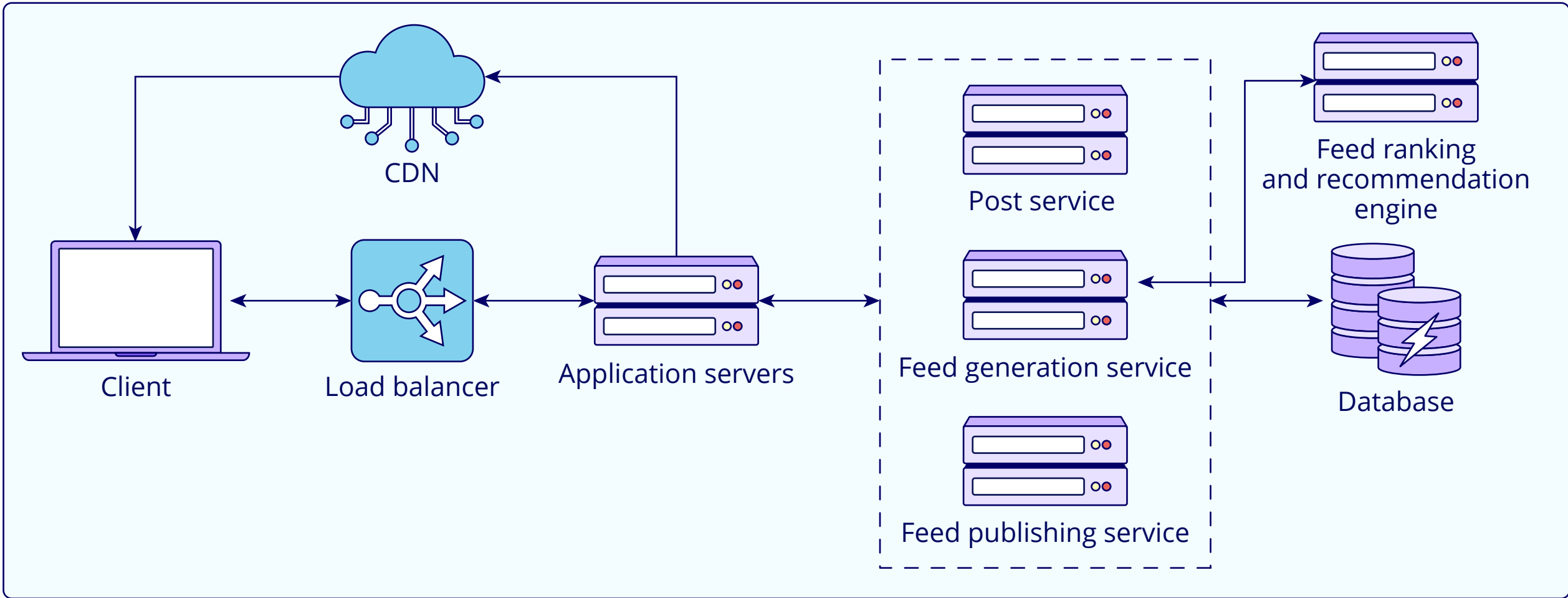


Functional requirements:

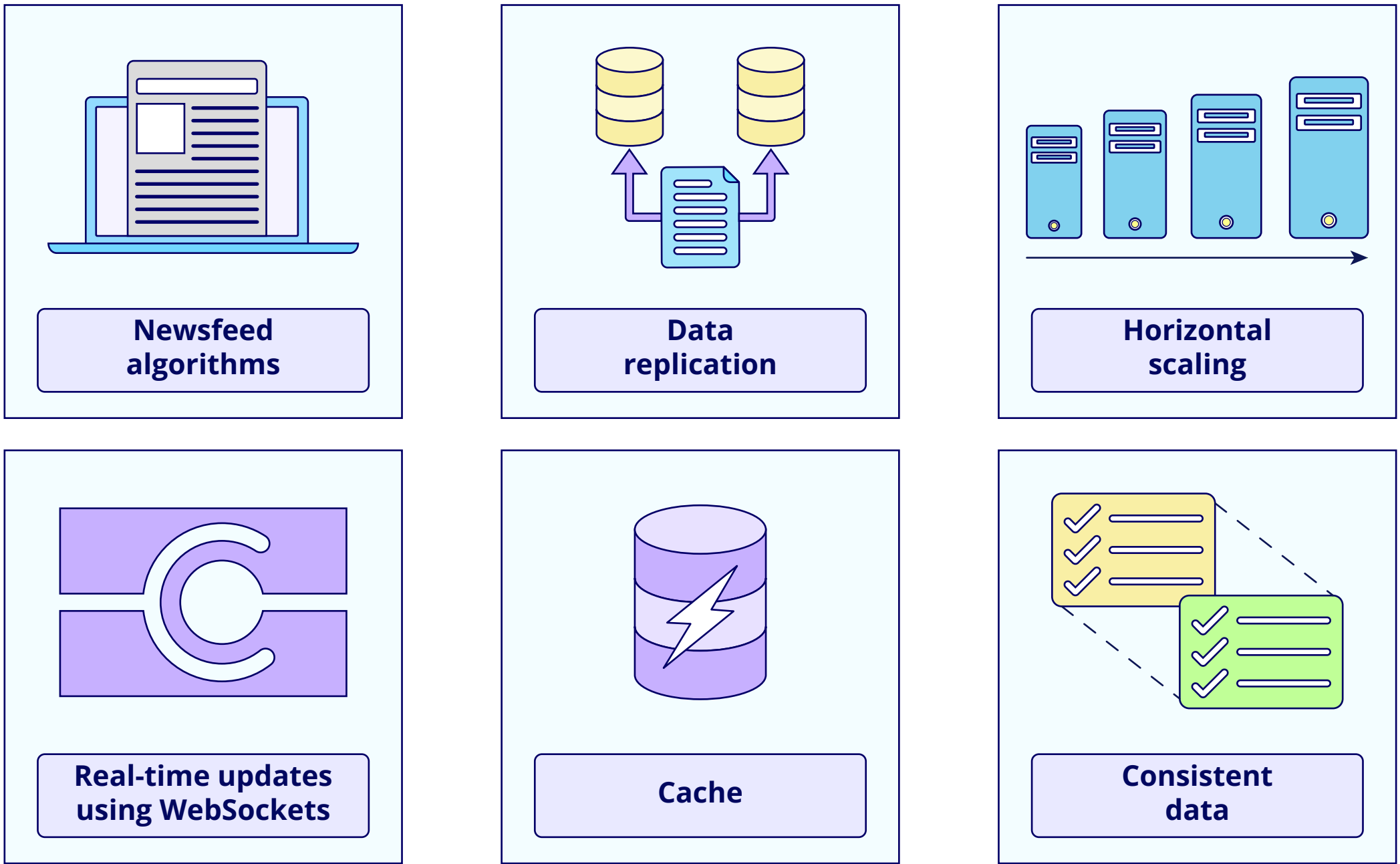
- Create a post
- Delete a post
- Edit a post
- Share a post
- Follow and unfollow users
- Search content
- View the system’s generated feed
- Like and dislike content

Nonfunctional requirements:

- Scalability
- Availability
- Low latency
- Durability
- Reliability



Key techniques to leverage



Cloud-Based Collaborative Editing Systems

Cloud-based collaborative apps


Google Drive


Dropbox

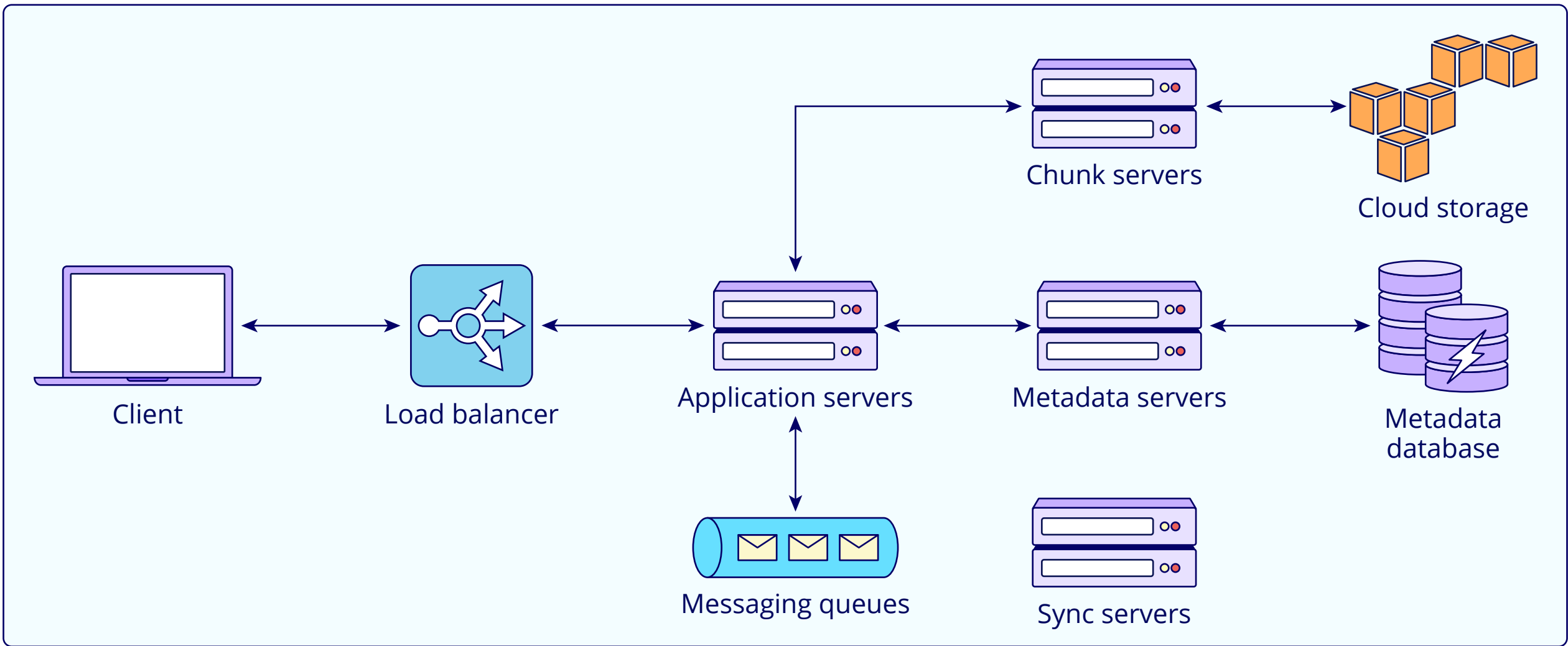

Microsoft 365

Functional requirements:

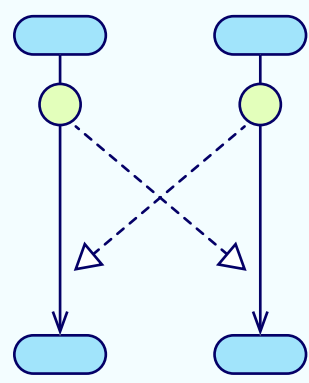
- Upload a document
- Edit documents
- Download documents
- Share documents
- Create and delete directories
- Documents synchronization

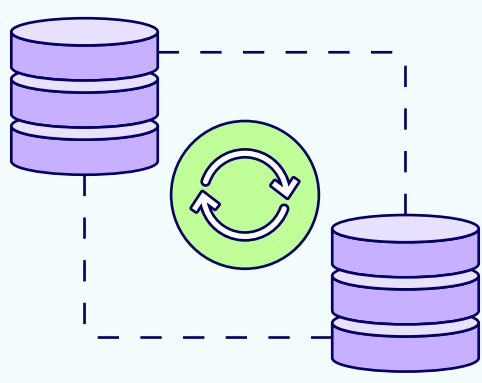
Nonfunctional requirements:

- Consistency
- Durability
- Reliability
- Availability
- Scalability
- Security



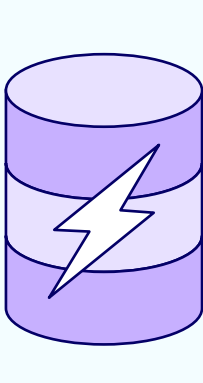
Key techniques to leverage

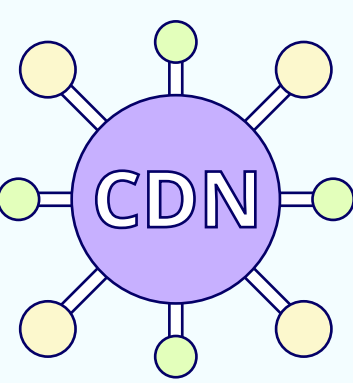

Operational transformation (OT)


Differential synchronization (DS)


Versioning


Access control

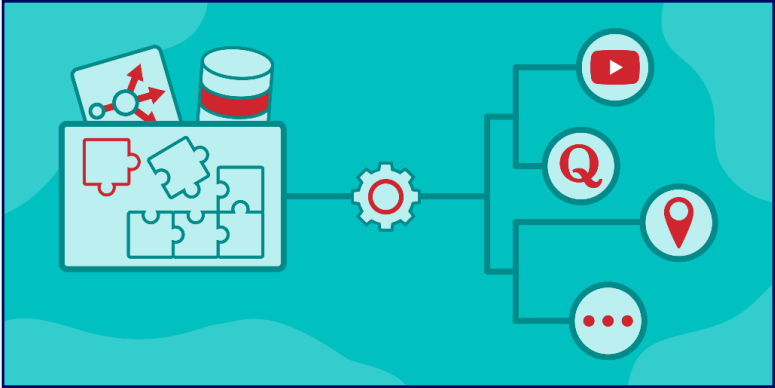

Cache


CDNs for delivery

Systems Design Courses

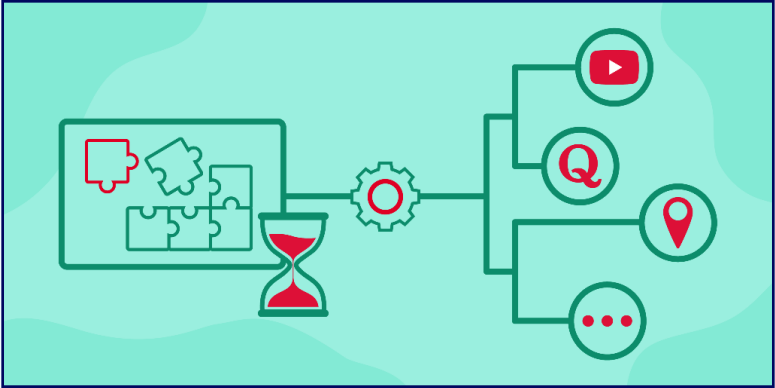
To learn more about System Design, you can take help from the following resources:

Grokking the Modern System Design Interview



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System Design Interview Prep Crash Course



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Grokking the Principles and Practices of Advanced System Design



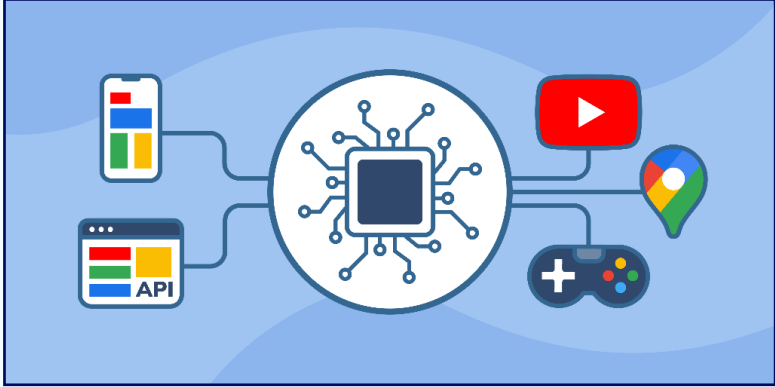
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Advanced System Design Interview Prep: Crash Course



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Grokking the Product Architecture Design Interview



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